VICTORY 2027

TRANSPIRATION

LECTURE-3 BIOLOGY

Upasana Ma'am



TOPICS TO BE COVERED

- Transpiration types
- Factors affecting Transpiration


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Q Wilting of plants during midday, despite adequate soil water is mainly due to:

A) Root pressure exceeding transpirational pull

B) Rate of transpiration exceeding water absorption

C) Permanent plasmolysis in guard cell

D) Increased Osmotic potential

Q. In tall trees, which ~~water~~ force is responsible for pulling water and minerals from the soil?

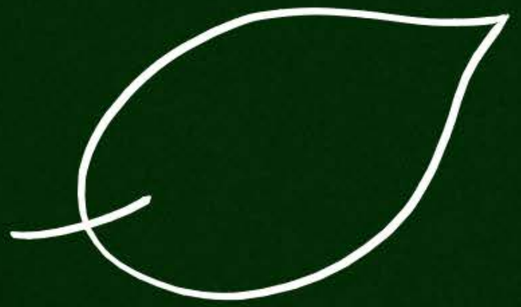
A) Gravitational force

B) Transportation force

C) Suction force / Transpirational pull ✓

[Monocot leaf]

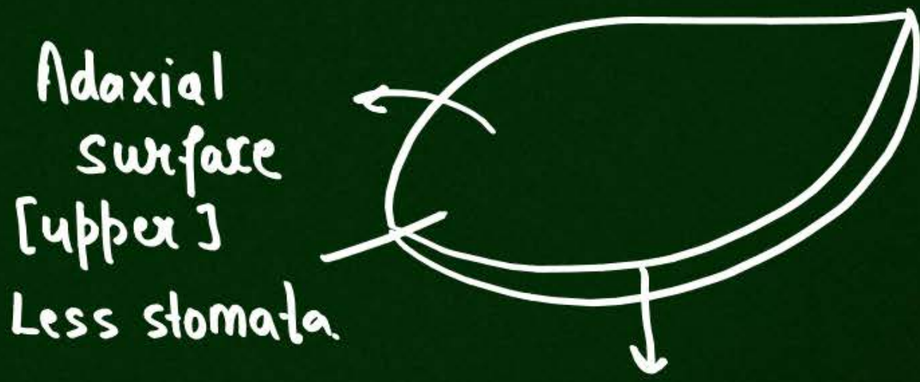
Maize



Has same number of stomata on both upper & lower surface

[Dicot leaf]

chick pea



Abaxial surface
↳ Below
More no of stomata.

Q. Transpiration is regulated by the movement of -

A) Guard Cells ✓

B) Palisade Cells

C) Epithelial Cells

D) None

CUTICULAR TRANSPIRATION

7-9%



Cuticle is a waxy layer secreted by the epidermis on the two surfaces of the leaf.



The greater the thickness of the cuticle, the lesser is the evaporation (transpiration).

Xerophytes



Desert plants tend to have thicker cuticles to cut down transpiration.





LENTICULAR TRANSPIRATION

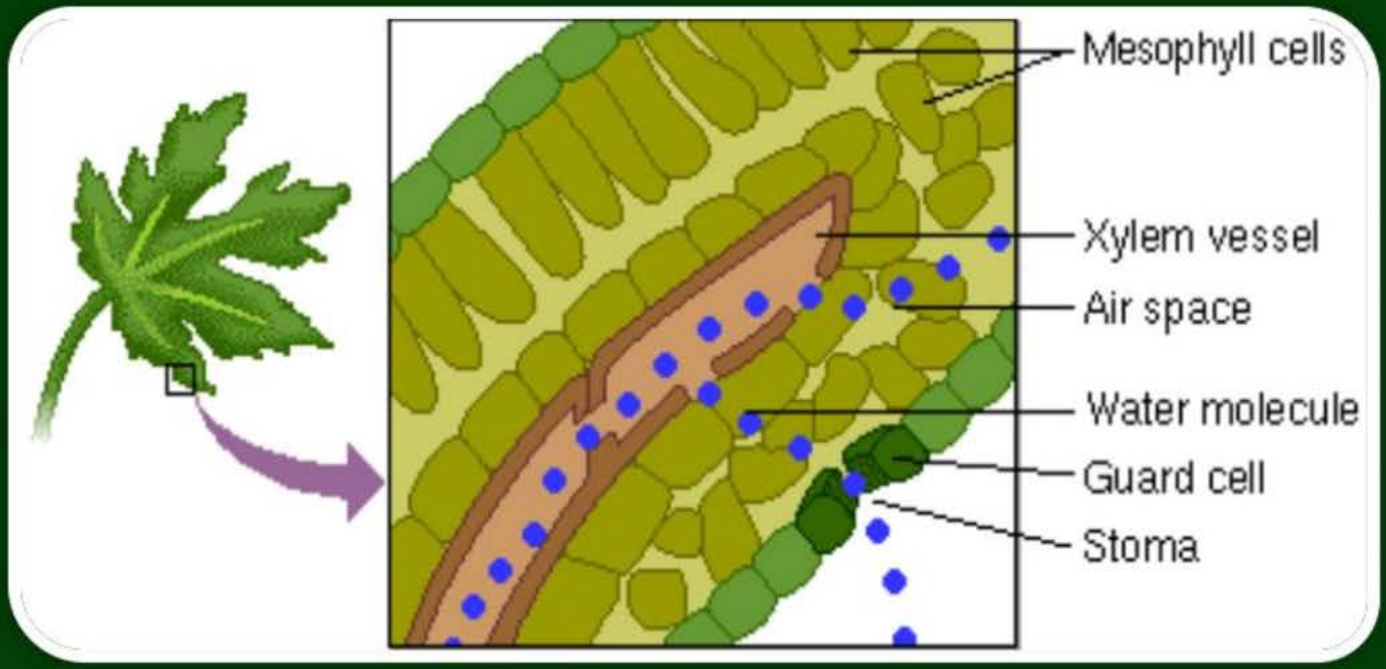
1-2%

- Lenticels are special openings that develop on the barks of older stems.
- These allow diffusion of gases for respiration.
- Lenticels never close. They remain open all the time.
- Water from the cell surfaces directly facing the lenticel evaporates and contributes to transpiration



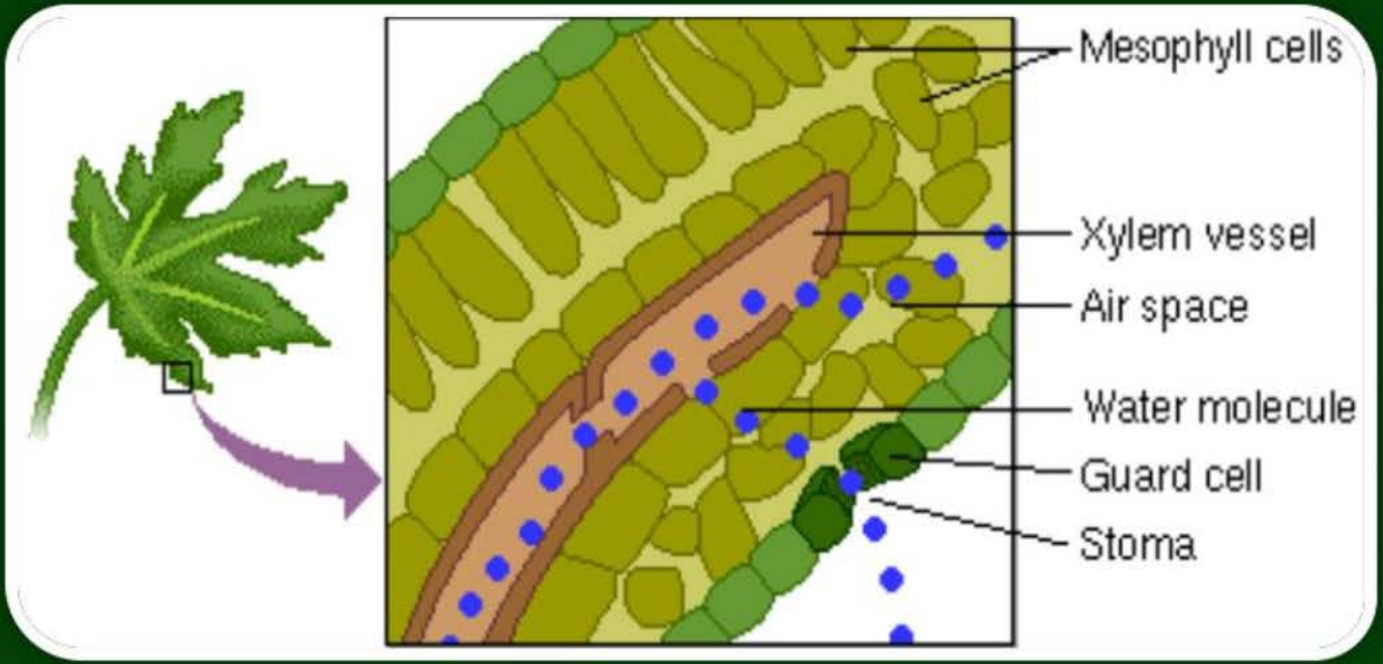
MECHANISM OF STOMATAL TRANSPIRATION

In stomatal transpiration, the water vapour escapes through the stomata of the leaf. Water, after absorption by the roots from the soil, rises up through the stem and reaches the tissues of the leaves through veins.

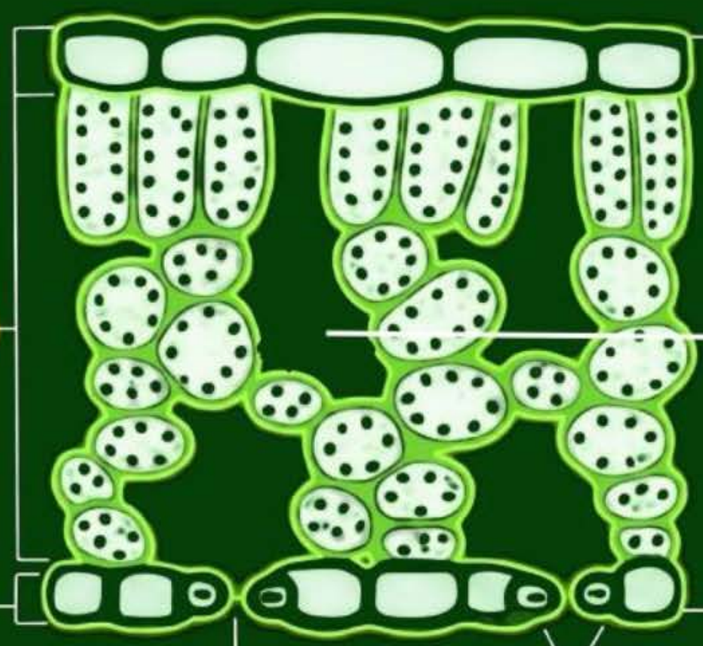


MECHANISM OF STOMATAL TRANSPIRATION

The water vapour formed first saturates the air in the intercellular spaces and then diffuses to the sub-stomatal space and finally through stomata.



MECHANISM OF STOMATAL TRANSPIRATION



Upper Epidermis

1 Cuticle
Prevents water loss

Mesophyll
2
Site of photosynthesis

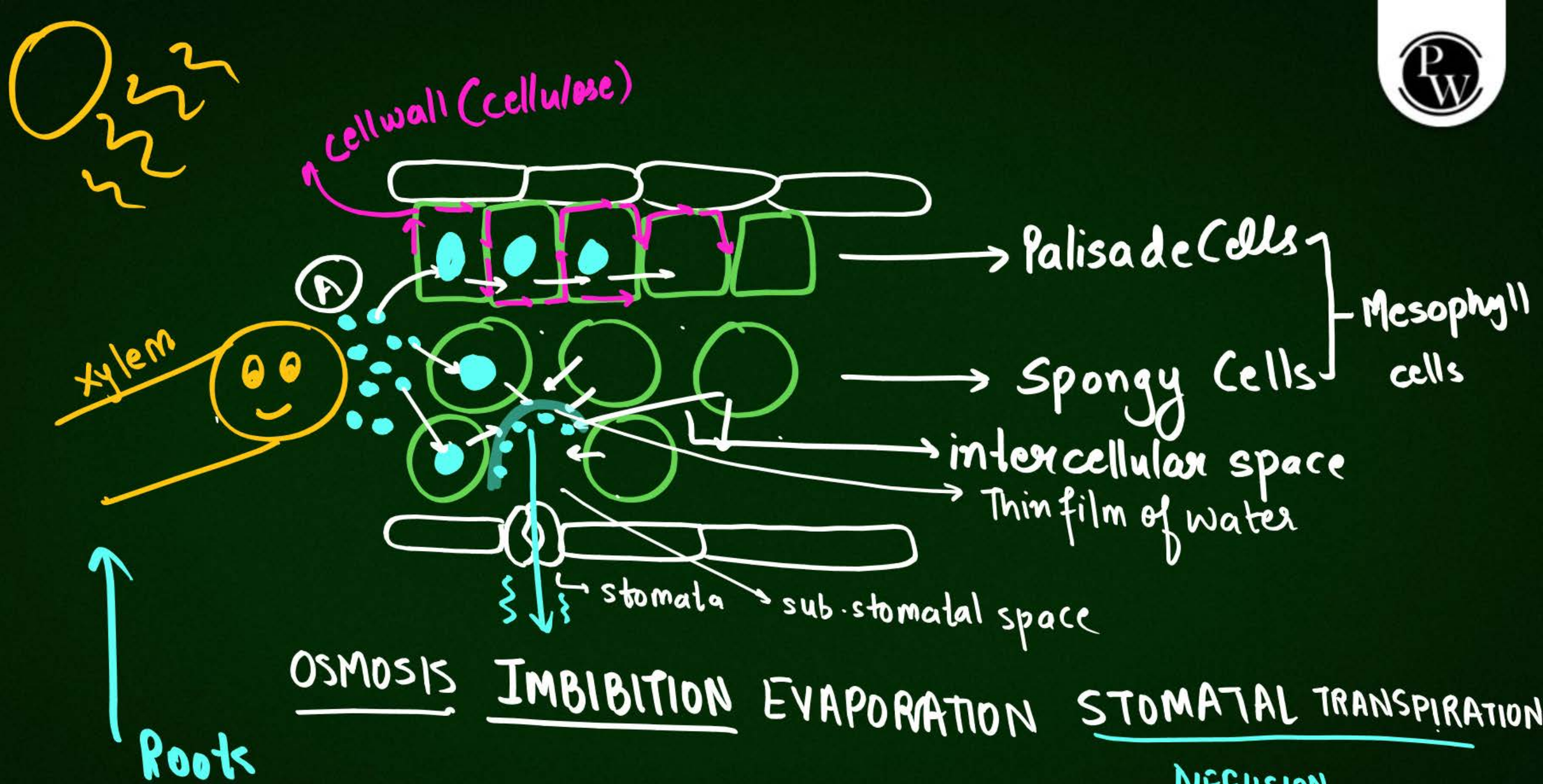
Air Space

Lower Epidermis

1 Cuticle

Stomata
3
Openings allow gases and water to move in and out of leaf

4 Guard cells
Open and close the stomata



OSMOSIS IMBIBITION EVAPORATION STOMATAL TRANSPIRATION

DIFFUSION

QUESTIONS

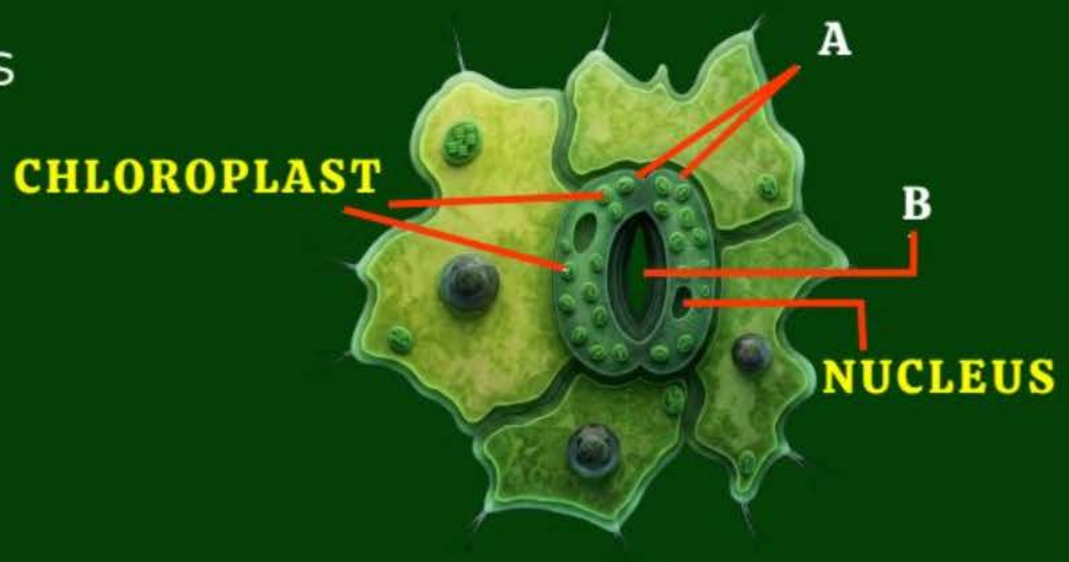
Label the guidelines A and B

A. Epidermal cells and guard cells

B. Guard cells and stoma

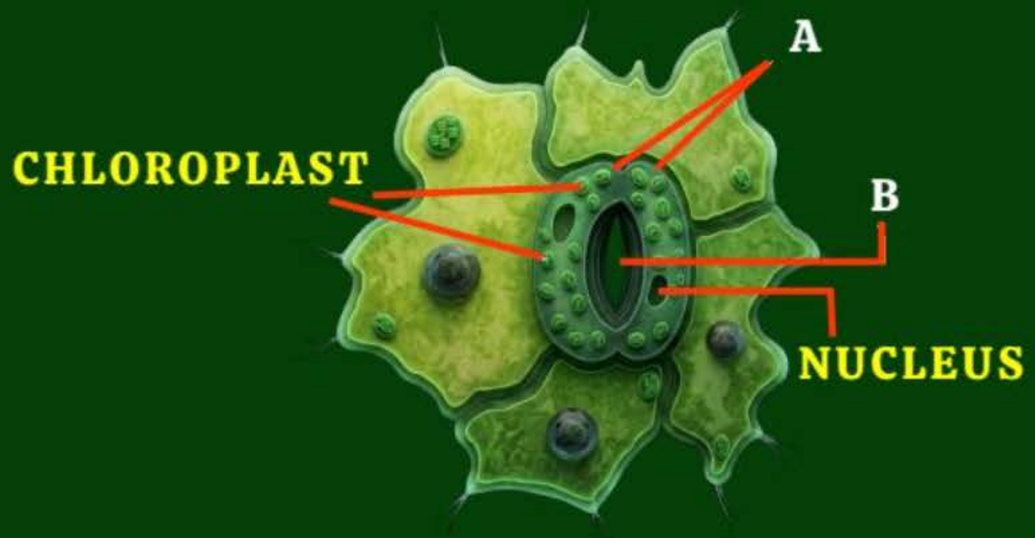
C. Stoma and guard cells

D. Epidermal cells



QUESTIONS

Mention the important role of A

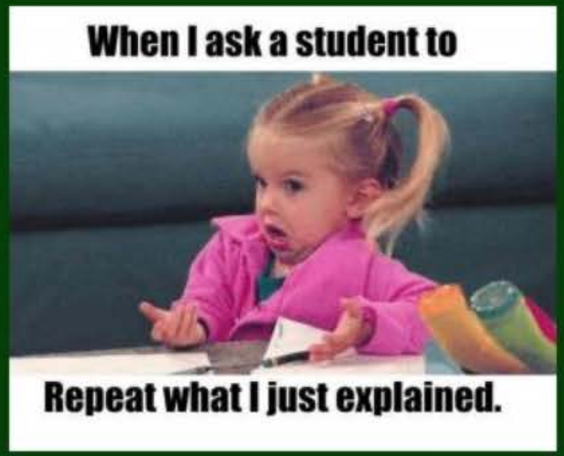


A Helps in photosynthesis

B Opening and closing of stoma

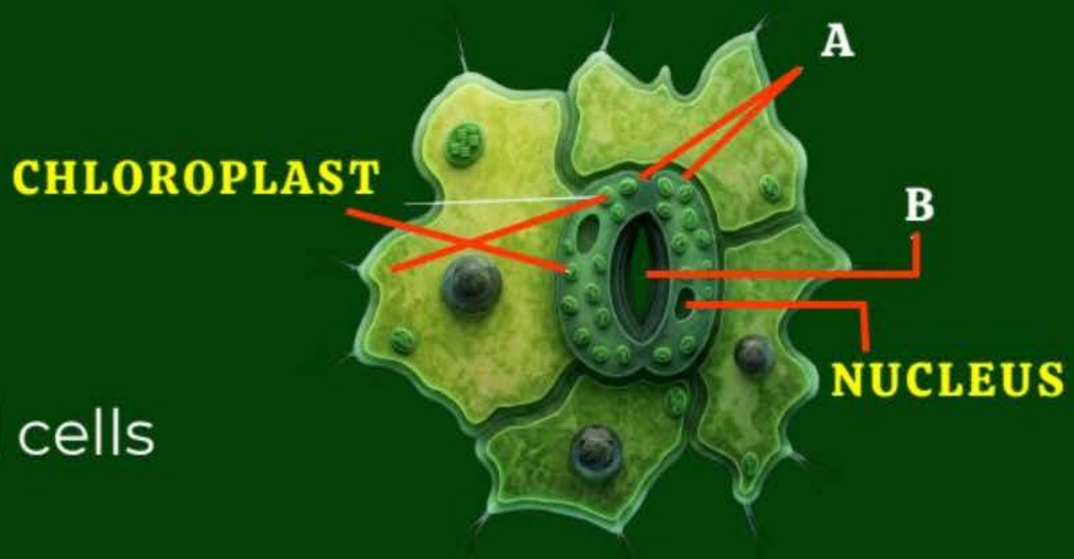
C Transpiration

D All of these



QUESTIONS

Mention the important role of B



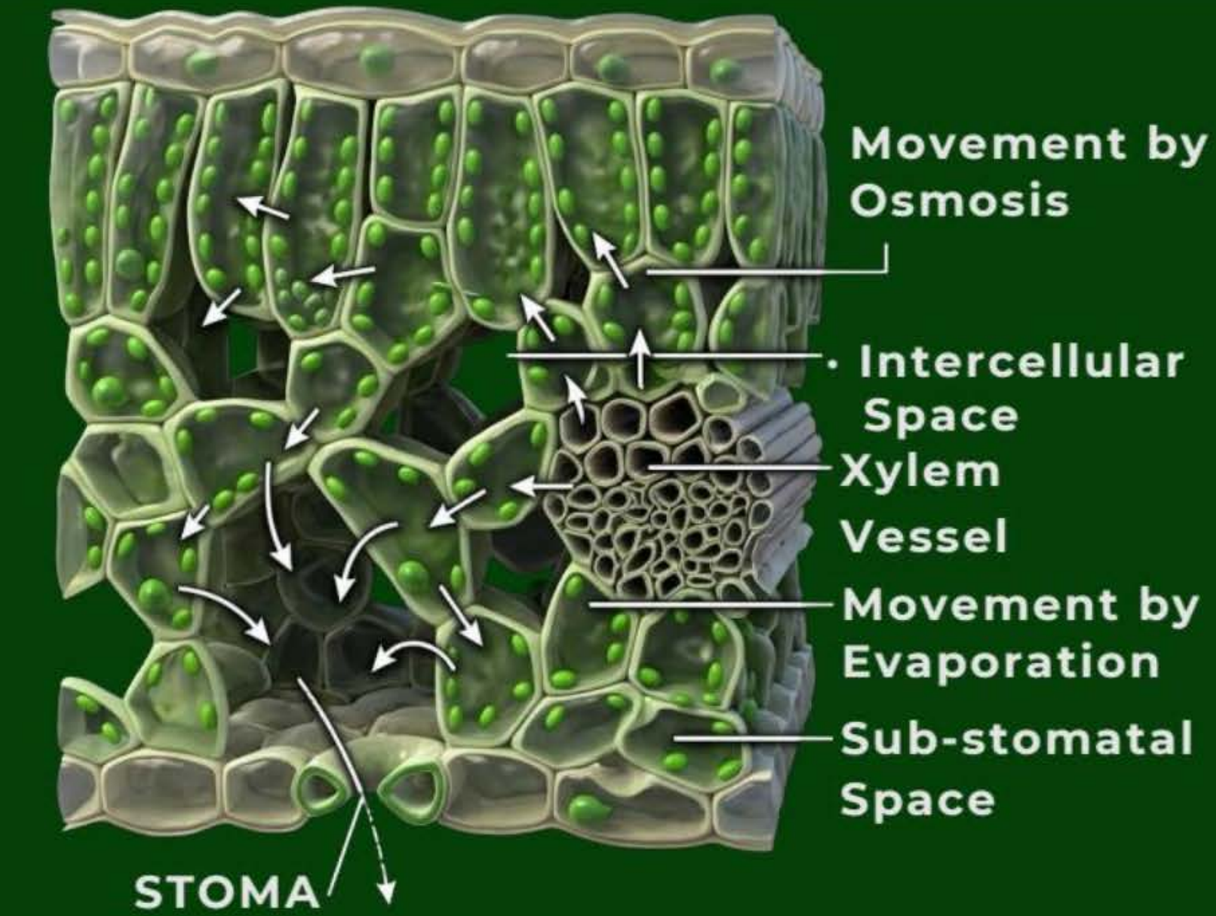
A. Helps in photosynthesis

B. Opening and closing of guard cells

C. Transpiration

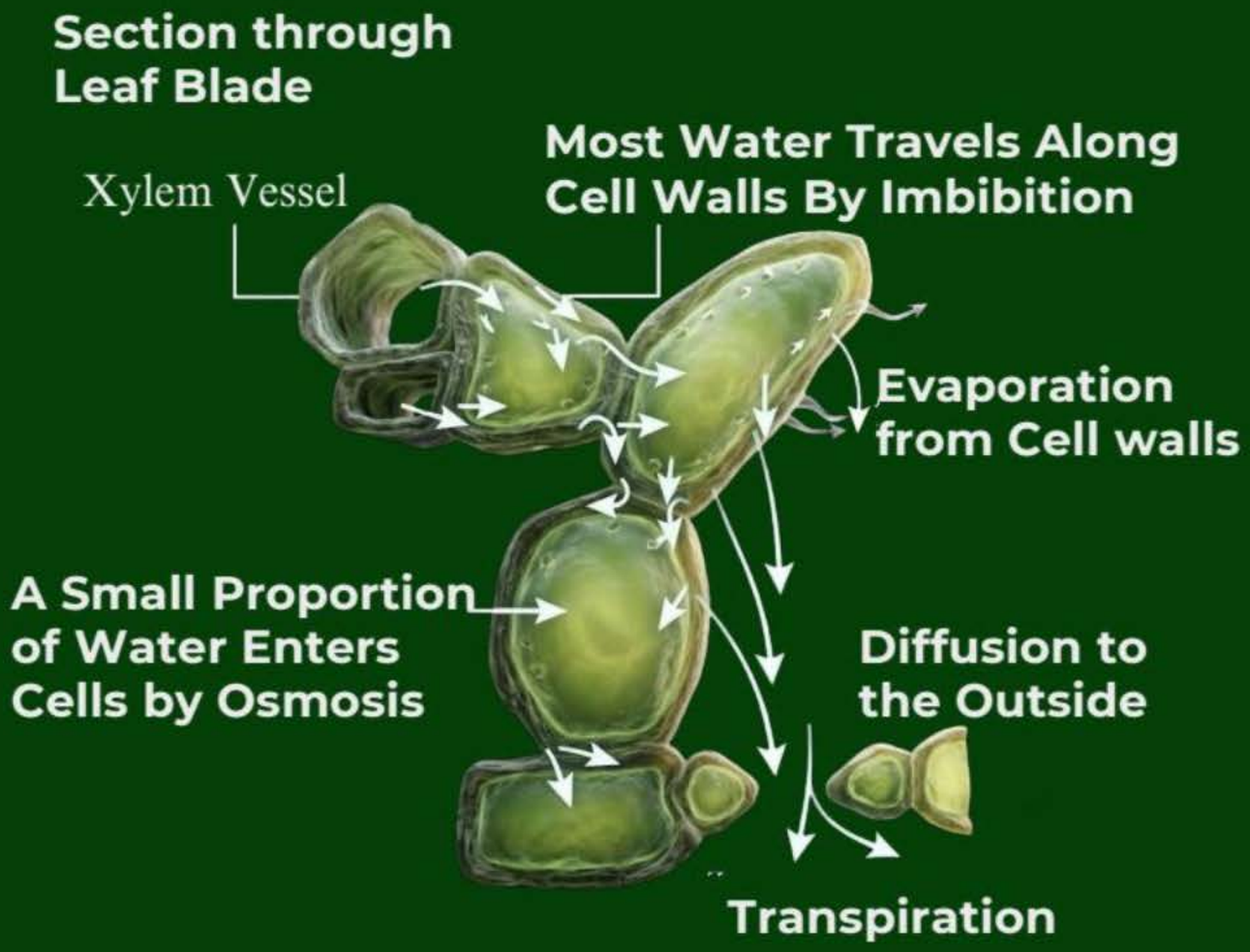
D. Exchange of gases

MECHANISM OF STOMATAL TRANSPIRATION



A Section through Leaf Blade

MECHANISM OF STOMATAL TRANSPIRATION



DIFFERENCES BETWEEN STOMATA AND LENTICLES ON THE BASIS OF STRUCTURE, LOCATION AND FUNCTION.

STOMATA	LENTICELS
These are small openings surrounded by two bean-shaped guard cells. They can be opened and closed.	These are special openings which are not surrounded by guard cells. They remain open all the time.
They are mostly found on the abaxial surface of leaves. [Dicot leaf]	These are found on the surface of older woody stems.
These are responsible for diffusion of gases during photosynthesis and respiration. They also help in transpiration.	These are mainly responsible for diffusion of gases during respiration. They also help in transpiration.

Exudation

The oozing out of water from parts of plant either in the form of water [Gutation] or when the plant is injured [Bleeding].

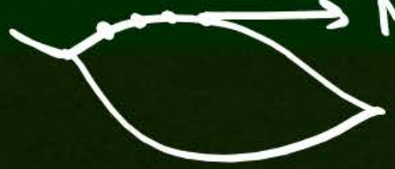
DIFFERENCES BETWEEN GUTTATION AND BLEEDING

GUTTATION	BLEEDING
It occurs from the edges of leaves by hydathodes in uninjured plants.	It occurs from any cut or injured part of a plant.
The exudate is mainly water with some dissolved mineral salts.	The exudate is mainly plant sap and sugars.
It occurs during early mornings or late nights.	It occurs at the time of injury.
It happens in certain plants like banana, Nasturtium strawberry.	It occurs in all plants that have been cut or injured.

Transpiration	Guttation
Water is lost in the form of water vapours.	_____ water droplets.
It occurs through Stomata, Cuticle, lenticels	_____ Hydathodes.
It is controlled by different factors.	It occurs when plants have extra water. for eg. in Rainy Season.

Guttation





Margin openings

↓

HYDATHODES

Bleeding ✓



↳ cell sap

[solution in plant cell]

(Water + solutes)



FACTORS THAT AFFECT TRANSPIRATION

A. External factors

V. V imp.

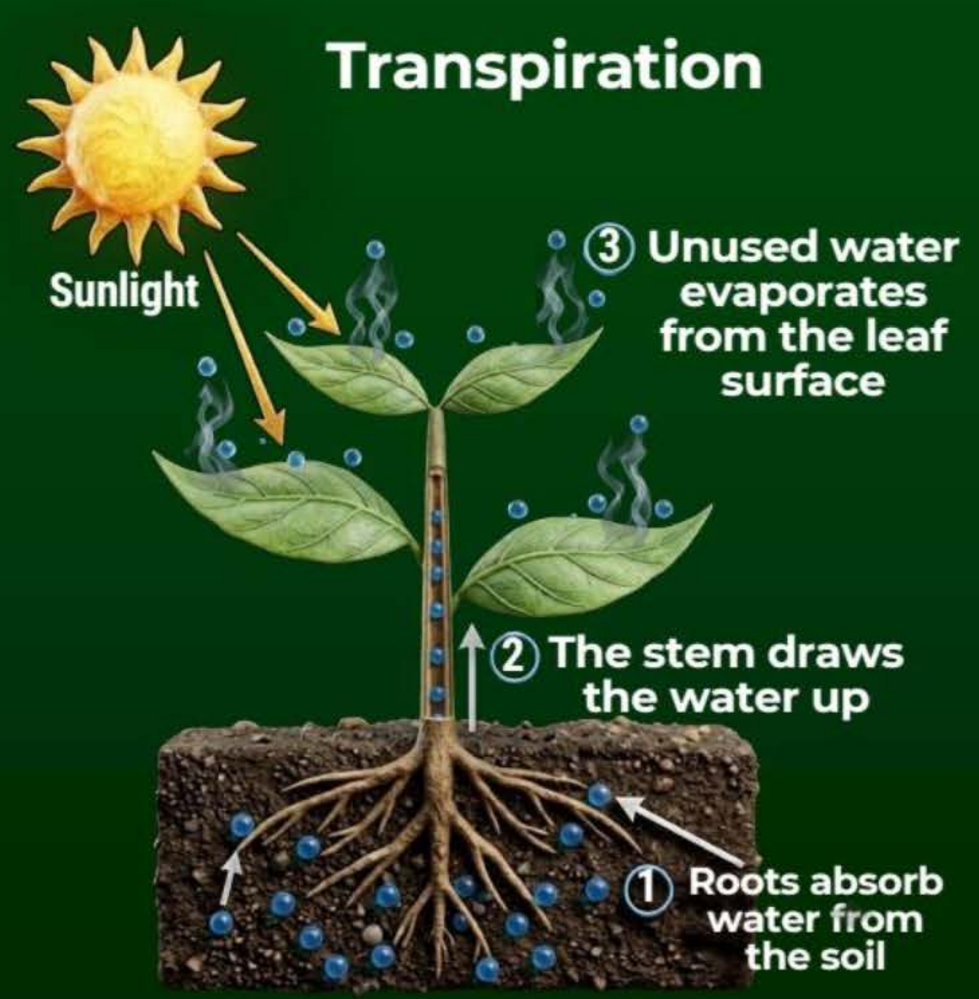
PYQs

2025

2026

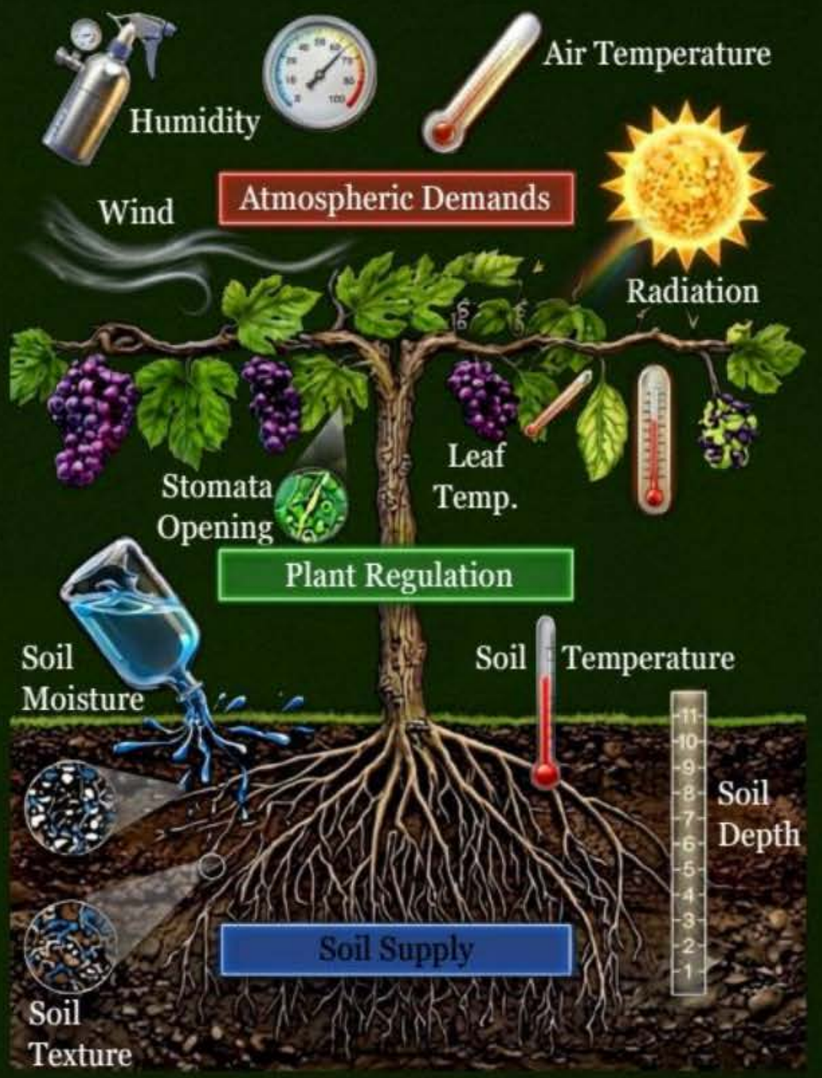
INTENSITY OF SUNLIGHT

- During the day, the stomata are open to facilitate the inward diffusion of CO_2 for photosynthesis.
- At night they are closed. Therefore, more transpiration occurs during the day.
- When it is cloudy during the day, the stomata are partially closed and transpiration is reduced.



TEMPERATURE

- If the outside temperature is higher, there is more evaporation from the leaves, therefore, more transpiration.
- Increase in temperature allows more water to evaporate and the decrease in temperature reduces evaporation.
- Warm air can hold more water than cold air.



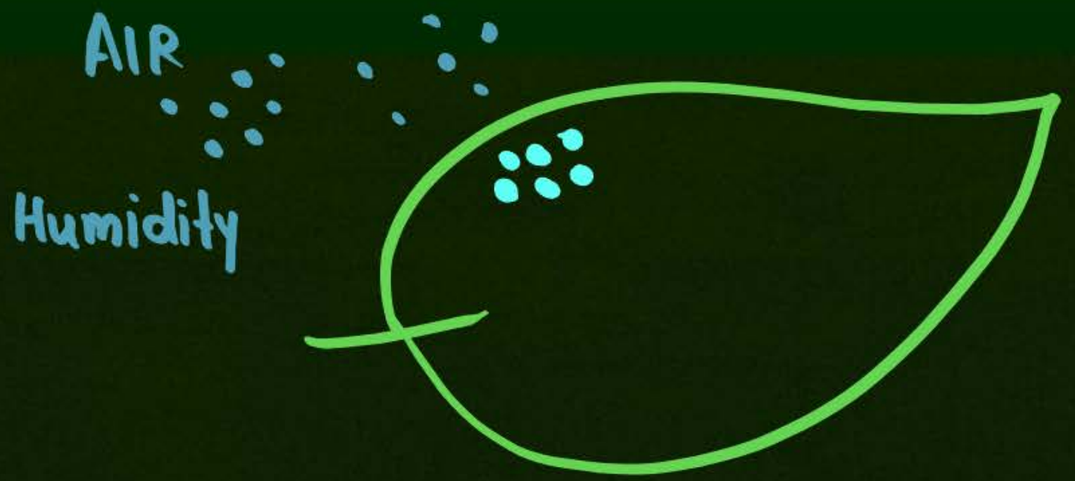
VELOCITY OF WIND

- Transpiration increases with the velocity of wind. If the wind blows faster, the water vapour released during transpiration is removed faster and the area outside the leaf does not get saturated with water vapour.



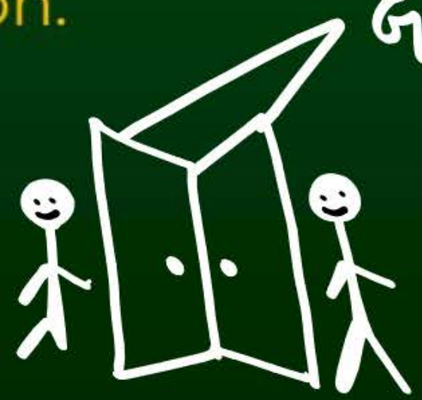
HUMIDITY

- Transpiration is reduced if the air outside is humid.
- High humidity in the air reduces the rate of outward diffusion of the internal water vapour across stomata, thereby reducing the rate of transpiration.



CARBON DIOXIDE (CO_2)

- Increase in the CO_2 level in the outside air over normal 0.03% causes stomatal closure and results in the decrease of transpiration.



CO_2 (Normal) 0.03%.

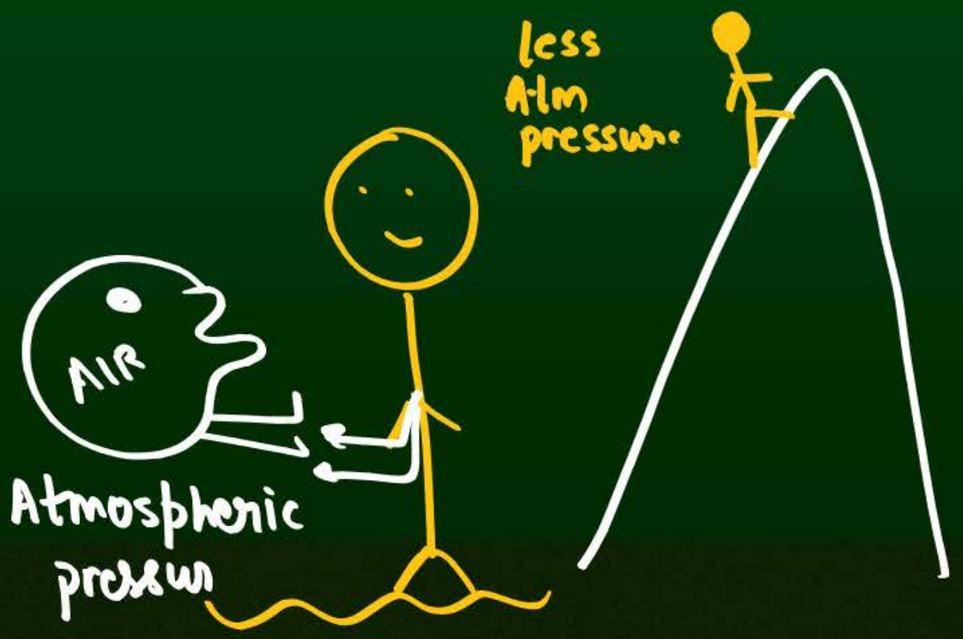


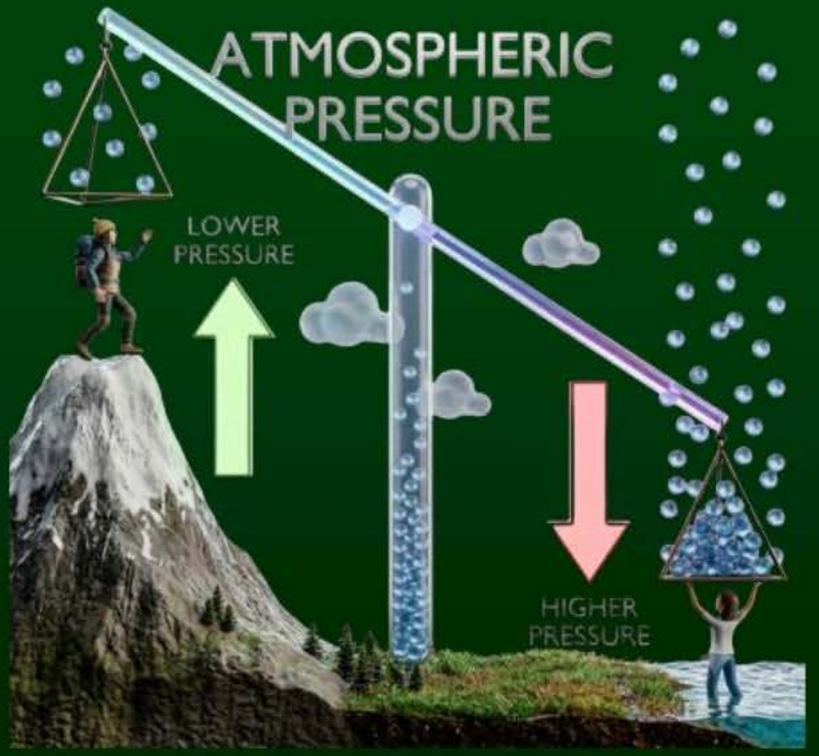
If CO_2 reaches 0.05%↑
Stomata Closed.
Reduced Transpiration

ATMOSPHERIC PRESSURE

- Rate of transpiration increases with the decrease in atmospheric pressure. Thus, it enhances diffusion of water vapour.

At Heights, more transpiration occur, because of less atm pressure.

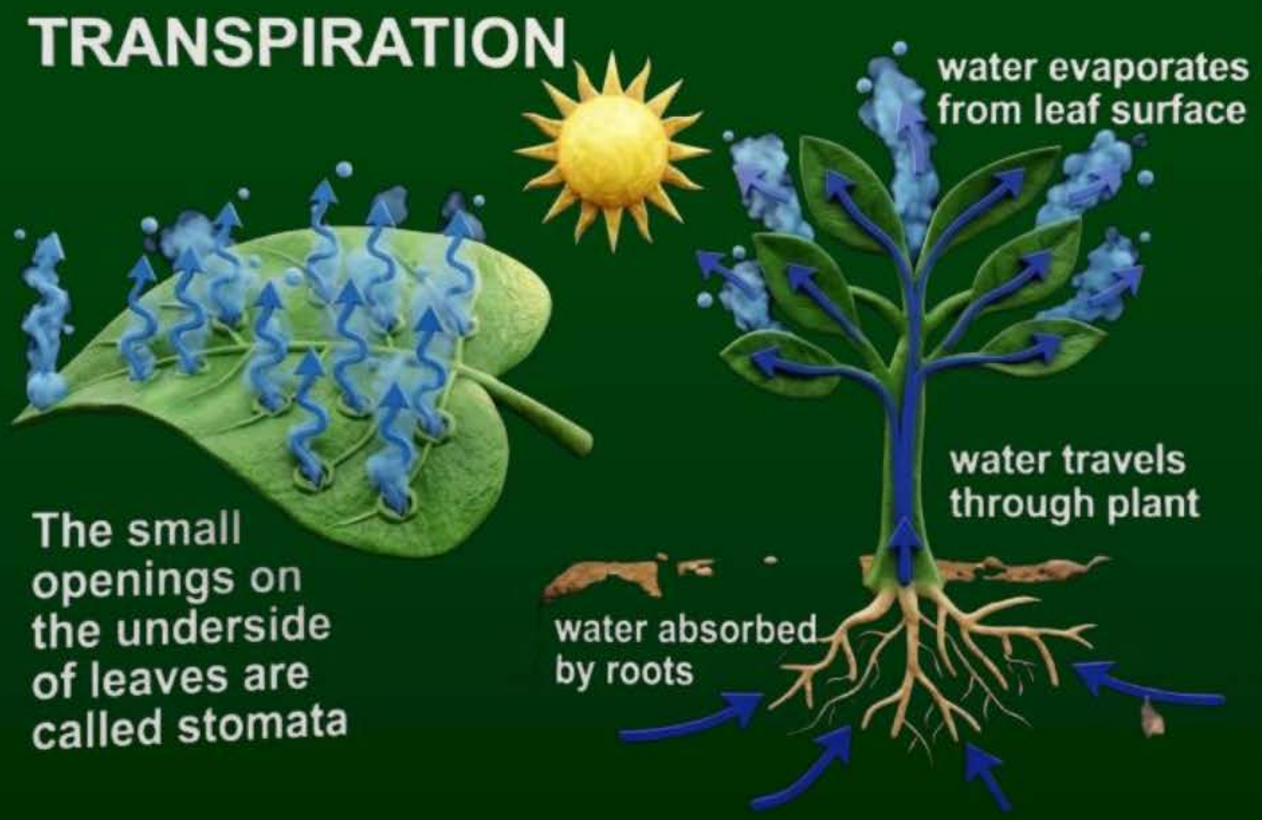




INTERNAL FACTOR

Water content of the leaves:

- If the water content of the leaves decreases due to insufficient absorption of water by the roots, the leaves wilt and transpiration is reduced. Such reduction in transpiration is indirectly due to the closure of stomata and it is a natural mechanism of conserving water within the plant.



INTERNAL FACTOR

Transpiration losses alot of plant's water and also create transpiration pull for water absorption



Maarenge bhi hum aur bachaayenge bhi hum

Adaptations in plants to reduce excessive transpiration



e.g Xerophytes

1) **Sunken Stomata**

stomata sunken or covered by hair.
e.g Nereium



2) **Fewer Stomata**

Number of stomata may be reduced

3) **Reduced Exposed surfaces**

leaves become wavy, rolled or folded to reduce surface area

4) **Narrow leaves**

Narrow leaves to reduce the surface area e.g. Nereium



5) **Loss of leaves**

leaves absent or changed into spines.

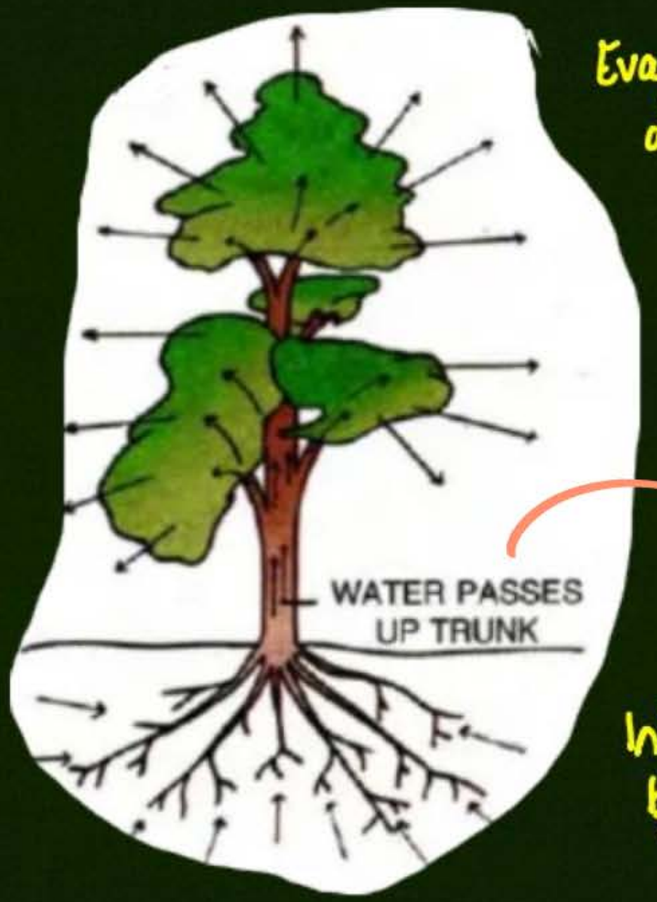


6) **Thick cuticle**

Leaves covered with thick cuticle. e.g. Banyan & evergreen trees.

THE TRANSPIRATION STREAM

Diagram Based Question

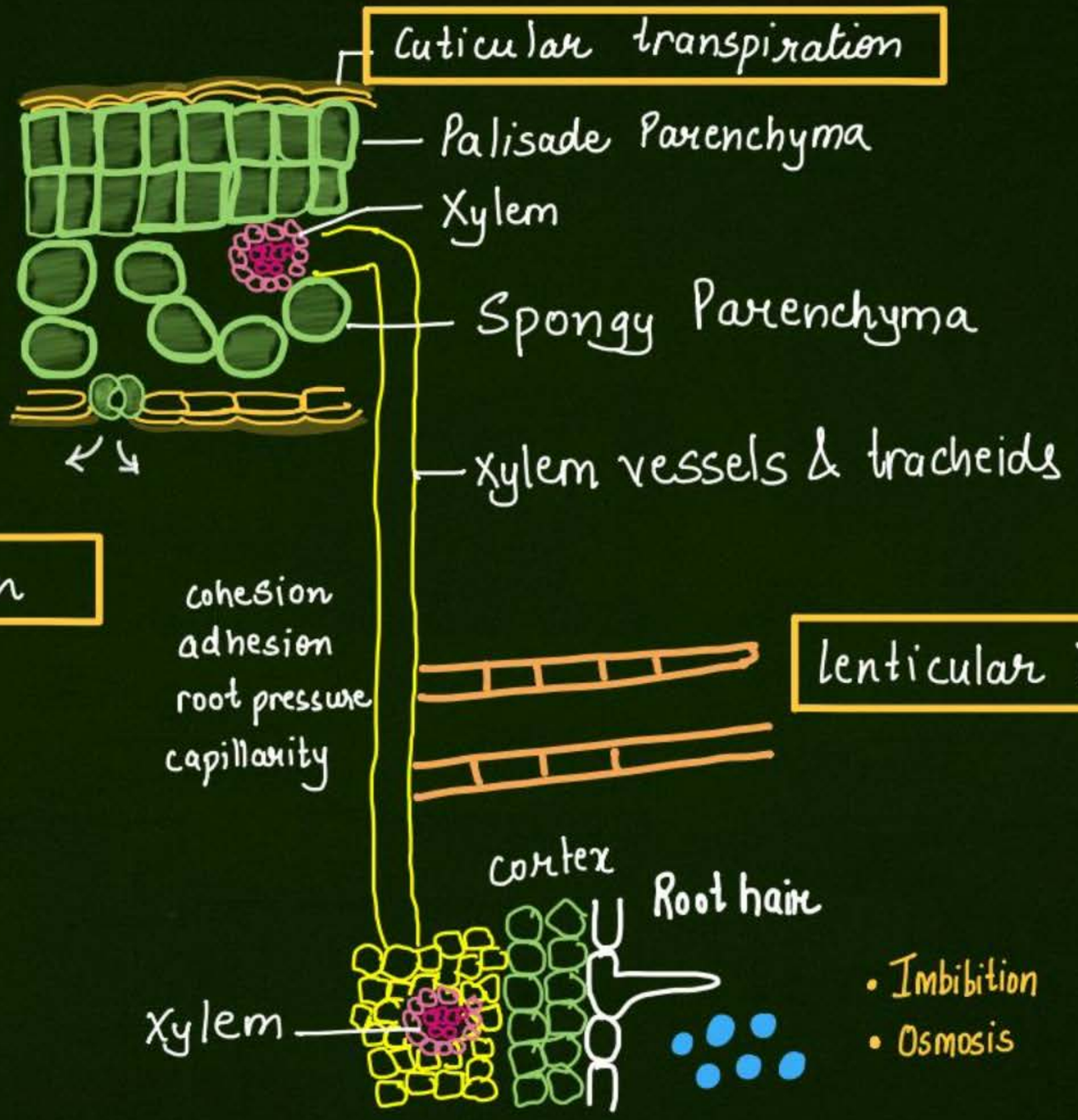


Evaporation into atmosphere from leaf surface (C) Transpiration

(B) Ascent of Sap

Water absorbed by roots from soil (A) Root pressure

Summary


Cuticular transpiration
Palisade Parenchyma
Xylem
Spongy Parenchyma
Xylem vessels & tracheids
Stomatal Transpiration
cohesion
adhesion
root pressure
capillarity
lenticular Transpiration
cortex
Root hair
Xylem
• Imbibition
• Osmosis

Thank You